

Mode 11 — Complex

$$\begin{aligned} \text{C-DP16 } (8 \times 8) &\rightarrow \text{DP32 } (8 \times 8) \rightarrow \text{DP16 } (8 \times 8) \rightarrow \text{DP16 } (8 \times 4) \\ &\rightarrow \text{DP8 } (8 \times 4) \end{aligned}$$

Step 1 — C-DP16 (8×8)

$$\text{C-DP16 } (8 \times 8) = D = \sum_{i=0}^{15} a_i b_i, \quad a_i = a_i^{re} + j a_i^{im}, \quad b_i = b_i^{re} + j b_i^{im}$$

Step 2 — DP32 (8×8)

$$\text{Re}(D) = \underbrace{\sum_{i=0}^{15} a_i^{re} b_i^{re} - \sum_{i=0}^{15} a_i^{im} b_i^{im}}_{\text{DP32 } (8 \times 8)}$$

$$\text{Im}(D) = \underbrace{\sum_{i=0}^{15} a_i^{re} b_i^{im} + \sum_{i=0}^{15} a_i^{im} b_i^{re}}_{\text{DP32 } (8 \times 8)}$$

Step 3 — DP16 (8×8)

$$\text{Re}(D) = \underbrace{\sum_{i=0}^{15} a_i^{re} b_i^{re}}_{\text{DP16 } (8 \times 8)} - \underbrace{\sum_{i=0}^{15} a_i^{im} b_i^{im}}_{\text{DP16 } (8 \times 8)}$$

$$\text{Im}(D) = \underbrace{\sum_{i=0}^{15} a_i^{re} b_i^{im}}_{\text{DP16 } (8 \times 8)} + \underbrace{\sum_{i=0}^{15} a_i^{im} b_i^{re}}_{\text{DP16 } (8 \times 8)}$$

Step 4 — DP16 (8×4) (nibble-split $b = 2^4 b^{hi} + b^{lo}$)

$$\text{Re}(D) = 2^4 \underbrace{\sum_{i=0}^{15} a_i^{re} b_i^{re,hi}}_{\text{DP16 } (8 \times 4)} + \underbrace{\sum_{i=0}^{15} a_i^{re} b_i^{re,lo}}_{\text{DP16 } (8 \times 4)} - 2^4 \underbrace{\sum_{i=0}^{15} a_i^{im} b_i^{im,hi}}_{\text{DP16 } (8 \times 4)} - \underbrace{\sum_{i=0}^{15} a_i^{im} b_i^{im,lo}}_{\text{DP16 } (8 \times 4)}$$

$$\text{Im}(D) = 2^4 \underbrace{\sum_{i=0}^{15} a_i^{re} b_i^{im,hi}}_{\text{DP16 } (8 \times 4)} + \underbrace{\sum_{i=0}^{15} a_i^{re} b_i^{im,lo}}_{\text{DP16 } (8 \times 4)} + 2^4 \underbrace{\sum_{i=0}^{15} a_i^{im} b_i^{re,hi}}_{\text{DP16 } (8 \times 4)} + \underbrace{\sum_{i=0}^{15} a_i^{im} b_i^{re,lo}}_{\text{DP16 } (8 \times 4)}$$

Step 5 — DP8 (8×4) (index-split $[0, 15] = [0, 7] \cup [8, 15]$)

$$\begin{aligned}
\text{Re}(D) &= 2^4 \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{re,hi}}_{\text{DP8 (8×4)}} + 2^4 \underbrace{\sum_{i=8}^{15} a_i^{re} b_i^{re,hi}}_{\text{DP8 (8×4)}} + \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{re,lo}}_{\text{DP8 (8×4)}} + \underbrace{\sum_{i=8}^{15} a_i^{re} b_i^{re,lo}}_{\text{DP8 (8×4)}} \\
&\quad - 2^4 \underbrace{\sum_{i=0}^7 a_i^{im} b_i^{im,hi}}_{\text{DP8 (8×4)}} - 2^4 \underbrace{\sum_{i=8}^{15} a_i^{im} b_i^{im,hi}}_{\text{DP8 (8×4)}} - \underbrace{\sum_{i=0}^7 a_i^{im} b_i^{im,lo}}_{\text{DP8 (8×4)}} - \underbrace{\sum_{i=8}^{15} a_i^{im} b_i^{im,lo}}_{\text{DP8 (8×4)}} \\
\text{Im}(D) &= 2^4 \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{im,hi}}_{\text{DP8 (8×4)}} + 2^4 \underbrace{\sum_{i=8}^{15} a_i^{re} b_i^{im,hi}}_{\text{DP8 (8×4)}} + \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{im,lo}}_{\text{DP8 (8×4)}} + \underbrace{\sum_{i=8}^{15} a_i^{re} b_i^{im,lo}}_{\text{DP8 (8×4)}} \\
&\quad + 2^4 \underbrace{\sum_{i=0}^7 a_i^{im} b_i^{re,hi}}_{\text{DP8 (8×4)}} + 2^4 \underbrace{\sum_{i=8}^{15} a_i^{im} b_i^{re,hi}}_{\text{DP8 (8×4)}} + \underbrace{\sum_{i=0}^7 a_i^{im} b_i^{re,lo}}_{\text{DP8 (8×4)}} + \underbrace{\sum_{i=8}^{15} a_i^{im} b_i^{re,lo}}_{\text{DP8 (8×4)}}
\end{aligned}$$